

WACV 2027 Workshop Proposal

UrbanState: Multimodal Urban Sensing and Representation for Autonomous Mobility

Summary

Workshop Title	UrbanState: Multimodal Urban Sensing and Representation for Autonomous Mobility
Acronym	UrbanState
Edition	1st
Keywords	Multimodal perception, autonomous driving, city-scale representation, VLMs/LLMs, embodied AI, traffic sensing
Primary contact name and email	Dr. Reeshad Khan, reeshad.khan@ucf.edu
Half or Full Day	Half Day
Anticipated Audience Size	50–80
Requested Number of Poster Boards	12
Papers Published in Proceedings?	Yes

1 Tentative Call for Papers

UrbanState invites work on methods that transform multimodal, distributed urban observations into persistent, uncertainty-aware representations for perception, reasoning, prediction, and action in autonomous mobility system. We invite contributions in areas including:

- Multimodal and cross-platform perception for autonomous systems
- Urban scene representation and understanding
- Representation-informed perception, prediction, planning, and end-to-end mobility
- Cross-city/cross-sensor transfer; vision-language and multimodal foundation models
- Urban vision datasets, benchmarks, simulation, and evaluation
- Survey paper and other applications for urban sensing and representation

We plan to accept regular papers of 8 pages and extended abstracts of up to 4 pages, following the main conference’s guidelines. Regular papers will be included in the WACV 2027 proceedings.

2 Motivation, Impact, and Expected Outcomes

The central research gap is not a lack of urban sensing, but the absence of representations that persist and uncertainty-aware across platforms, locations, and time and that can support decisions by road users, robots and infrastructure. Existing research often treats sensors and mobility trace as separate inputs, limiting cross-platform fusion, transfer and city-scale reasoning.

UrbanState will bring together the computer vision, autonomous driving, intelligent transportation, and urban computing communities to advance cross-platform fusion, city-scale mapping, digital twins, uncertainty modeling, and representation-informed decision making. Expected outcomes include a shared research agenda, new cross-community collaborations, and clearer dataset, benchmark, and evaluation requirements for transferable urban representations. Orlando’s active AV testing environment and WACV’s applied focus make it a strong setting for discussion and collaboration.

3 Brief Bios of Organizers

- **Dr. Reeshad Khan**, University of Central Florida. reeshad.khan@ucf.edu, <https://reeshad-khan.github.io>. Dr. Khan is a Postdoctoral Scholar and he works on sensor co-design and end-to-end driving presented at ICCV, CVPR, and ECCV-affiliated workshops, including TinyBEV; currently works on vision-grounded urban mobility representation at UCF.
- **Dr. Shaurya Agarwal**, University of Central Florida. shaurya.agarwal@ucf.edu, <https://www.cecs.ucf.edu/sagarwal/> Dr. Agarwal is an Associate Professor at UCF and the founding director of the URBANITY Lab and director of the Future City Initiative, focusing his research on cyber-physical systems, smart transportation, and connected and autonomous vehicles. He has authored a book and published over 35 peer-reviewed papers, serves as an associate editor of IEEE Transactions on Intelligent Transportation Systems, and is a senior member of IEEE.
- **Dr. John Gauch**, University of Arkansas. jgauch@uark.edu, <http://csce.uark.edu/~jgauch>. Dr. Gauch is a Professor and his research spans real-time video analysis, image processing, and computer vision, including scene-transition detection, large-scale video indexing and retrieval, and biomedical image segmentation and motion tracking; he has published more than 70 papers.
- **Dr. Susan Gauch**, Professor, University of Arkansas. sgauch@uark.edu, <http://csce.uark.edu/~sgauch/>. Dr. Gauch is a Professor and her research covers information retrieval, ontologies, and personalization, including robust and personalized search engines and text mining; she has published more than 130 peer-reviewed papers, holds the Rodger S. Kline Leadership Chair, and previously served as head of the Department of Computer Science and Computer Engineering.
- **Dr. Chuheng Wei**, Purdue University. chuheng.wei@email.ucr.edu, <https://chuhengwei.com> Dr. Wei is a Postdoctoral Researcher and his research focuses on multimodal perception, cooperative sensing, and human-centered AI for autonomous driving and intelligent transportation systems. He has previously organized workshops at IV, ITSC, CVPR, and ICCV.
- **Dr. Ziran Wang**, Purdue University. ziran@purdue.edu, <https://ziranw.github.io> Dr. Wang is an Assistant Professor and he has published more than 80 papers and filed more than 50 patent applications on various topics of autonomous vehicles and intelligent transportation systems. He is an associate editor of IEEE Transactions on Intelligent Transportation Systems, Transactions on Intelligent Vehicles, IEEE Internet of Things Journal, and IEEE Network. He has previously organized workshops at CVPR, ICCV, WACV, ITSC, and IV.
- **Dr. Xishun Liao**, University of Central Florida. xishun.liao@ucf.edu, <https://xishunliao.github.io> Dr. Liao is an Assistant Professor and he has published more than 30 peer reviewed papers on intelligent transportation systems, digital twin technology, autonomous driving, and applied AI/ML for smart mobility solutions. He is an associate editor of the IEEE Open Journal of Intelligent Transportation Systems, and has previously organized workshops at ITSC and IV.

4 Tentative Program Committee

- Hao Cheng, Jintao Lao, Haoyan Wang, University of Central Florida
- Mezbaur Rahman Uschas, Tasnuva Tanjum Ferdous, University of Illinois Chicago
- Ibsa Kumara Jalata, Quantum Computing Inc.
- Vuong Ho, University of Texas at Dallas

Additional reviewers will be recruited to cover all CFP topics and submitted papers.

5 Format and Preliminary Schedule

Format: Half day. **Preliminary Schedule (tentative):**

8:00 Registration Opens **8:30–8:40** Opening Remarks
8:40–9:30 Keynote
9:30–10:00 Spotlight Paper 1
10:00–10:15 Coffee Break (aligned with main conference)
10:15–10:45 Invited Talk
10:45–11:15 Spotlight Paper 2
11:15–11:45 Invited Talk
11:45–12:00 Closing Remarks

6 Tentative Invited Speakers

- Dr. Mohan Trivedi, UC San Diego – Intelligent vehicles and multimodal transport sensing (tentative)
- Dr. Patrick Langechuan Liu, NVIDIA – Learning-based driving policies (tentative)
- Mashhour Solh, Waymo – Multimodal mapping and representation (tentative)
- Dr. Cathy Wu, MIT – Reinforcement learning for autonomous mobility (tentative)

7 Estimated Number of Submissions and Attendees

As a first edition and half-day workshop, we expect 12–20 submissions, based on comparable first-time workshops at recent CVPR, ICCV, and WACV editions. We estimate 50–80 attendees, drawn from WACV’s applied vision community, Central Florida’s AV research base, and Orlando’s AV industry.

8 Plans for Special Journal Issue or Book (if any)

Not for the 1st edition, but we will explore a special issue with a transportation journal in future editions.

9 Social Impact of the Workshop

Autonomous mobility is moving into shared, human-populated urban environments, raising the stakes on getting perception right for the people and infrastructure found in cities. Key stakeholders include road users, transport agencies, mobility providers, and communities affected by autonomous-system deployment. Our organizing team spans institutions and career stages, and the CFP encourages work on privacy-preserving sensing, fairness, accessibility, and uncertainty-aware evaluation. By bringing together researchers, industry practitioners, and public-sector stakeholders, the workshop aims to promote safer and more responsible deployment of autonomous mobility technologies.